



Backend Developer Coding Test

Project Title: Leave Management System (LMS)

Duration: 2 hours

Goal: Build a backend system to manage employee leave applications and basic approval.

Functional Requirements

1. Users

- Only two roles:
 - **Employee:** Can apply for leave.
 - **Manager:** Can approve or reject leave requests.

2. Leave Types

- Sick, Casual, and Earned.

3. Leave Application

- Employees can apply by submitting:
 - Leave Type
 - Start Date
 - End Date
 - Reason
- Prevent overlapping and backdated requests.

4. Leave Status

- A new leave request has status Pending.
- Manager can change status to Approved or Rejected with an optional comment.

5. Leave Balance

- Fixed annual quota per employee:
 - Sick (12), Casual (12), Earned (20)
- Leave requests should be blocked if balance is insufficient.

6. View Requests

- Employees can view their own leave requests.
 - Managers can view all employee leave requests with filters (status, type, date range).
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✅ Deliverables

- Implement the above functionality using any tools, frameworks, or architecture you prefer.
 - Use a database to store users, leaves, and balances.
 - Provide API documentation (Swagger).
 - Ensure validation, proper data handling, and clean code.
 - (Optional Bonus): Add basic authentication, or a simple audit trail for status changes.
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🎯 Notes

- You **may use AI tools** (e.g., ChatGPT, Copilot) for assistance.
 - Code should reflect your **own understanding and logic**.
 - Focus on clarity, correctness, and structure — not completeness of extras.
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📁 Submission

- Share your code via GitHub repo or as a ZIP.
 - Include a README.md with:
 - Setup instructions
 - API endpoint list
 - Any assumptions or notes
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